Fast Forward

The Future(s) of the Cinematic Arts

Holly Willis

WALLFLOWER PRESS
LONDON & NEW YORK

A Wallflower Press Book
Published by
Columbia University Press
Publishers Since 1893
New York Chichester, West Sussex
cup.columbia.edu

Copyright © 2016 Columbia University Press
All rights reserved
Wallflower Press® is a registered trademark of Columbia University Press

A complete CIP record is available from the Library of Congress

ISBN 978-0-231-17892-1 (cloth : alk. paper)
ISBN 978-0-231-17893-8 (pbk. : alk. paper)
ISBN 978-0-231-85097-1 (e-book)

Columbia University Press books are printed on permanent
and durable acid-free paper.
Printed in the United States of America

c 10 9 8 7 6 5 4 3 2 1
p 10 9 8 7 6 5 4 3 2 1

Cover design by Elsa Mathern

Chapter 3
Live Cinema

Perhaps one of the most striking embodiments of the post-cinematic/post-humanist conjunction occurs in a practice known as live cinema, in which artists dismantle and reconfigure the elements of the traditional cinematic apparatus in order to craft an event that straddles the boundary between mediated experience and live performance. Dubbed a 'beautifully located oxymoron' by Douglas Kahn (2011: 256), live cinema concretises the conflicting impulses of contemporary culture – the desire for presence in a world seemingly growing more and more evanescent, and the contradictory appeal for dispersal, divestiture and dissolution so much a part of networked existence. For Kahn, 'the oxymoron derives from the fact that cinema is a form of inanimate storage' and as such, live cinema becomes, oddly, a 'performance of a recording' (2011: 258). However, this conception of live cinema does little to illuminate a number of works that complicate liveness and which, I argue, work to limn the boundaries of the cinematic as well as existence itself within a networked culture. Live cinema specifically expands beyond the performance of a recording in order to query the relationship between live and recorded, and to muddy – or even refuse – the boundary between the two. Pushing a bit further, viewing this work within the context of a networked culture, we must note that the formerly distinct act of making a recording has been supplanted by the activity of designing the conditions for an ongoing and ambient form of processing. In place of optical media we have technical media, and in place of the 'things' of media past – videos, movies, DVDs and other objects – we have instead systems, networks, assemblages and environments. In short, in the context of computers and computation, we shift from objects to processes, and from notions of a singular identity to a kind of intra-subjectivity. Artists have responded by creating works that stage and enact the transition from one paradigm to another, and which call attention to the challenges we face in this upheaval. In this chapter, I will explore the work of four artists working in the realm of live cinema

with the goal of illustrating the issues raised as we make the – often uncomfortable – passage from one mode to another.

I define live cinema as the staging of a live event that includes the presentation of moving images and sounds for an audience with a performance component that calls attention to presence and liveness, as well as to the specificities of site, space and time. The images and sounds may be pre-recorded and mixed live, or generated in real time, but a core element within live cinema is that the project is not complete until it is performed, nor is it precisely repeatable. Indeed, despite the range of performance styles or genres gathered within this admittedly expansive definition, three aspects come to the foreground in solidifying a definition: first, the works insist on the live interaction of artwork, performer and audience and suggest a desire for the singularity and specificity of the event; second, they propose an interrogation of forms, asking us to contextualise the event within histories of cinema, video, performance, theatre, art and music, and to consider – or reconsider – the boundaries that we understand to delimit each form; and third, the works often play on the tension between the liquidity of information and the materiality of the things around us. They equivocate between the sense that we exist in a world characterised not by concrete spatial boundaries and fixed temporal coordinates, but instead by a mobile, accelerating experience of fluidity and flows, and yet we can feel the chairs on which we are seated during the performance. These three characteristics, then, hover at the boundary of cinema's specificity and its reconfiguration through digital media, as the materials specific to cinema – light, the projector, filmic emulsion – become mutable digital artefacts easily retooled for new uses.[1]

The current live cinema movement is presaged by earlier generations of artists who shared a similar desire to mix live performance and moving images in a body of work Gene Youngblood dubbed 'Expanded Cinema' in his book of the same title published in 1972.[2] Looking further back, as Paul Arthur notes in an introduction to a special issue on paracinema and performance in *Millennium Film Journal*, the earliest history of cinema, prior to its transformation into 'a thoroughly rationalized commodity' (2005: 3), included myriad formations of image-based experiences, including the use of fake railroad cars as the settings for Hale's photographic tours, as well as Abel Gance's three-screen design for the projection of *Napoleon* (1927). With the codification of the cinematic apparatus and the rise of the narrative feature film as cinema's primary object, experiments with live projection were relegated to the margins. However, this so-called marginal live cinema activity between the 1940s and 1990s was embraced by many avant-garde filmmakers, and indeed, Arthur offers a long list of participants who experimented with live projection performances: Jordan Belson, Harry Smith, Jack Smith, Barbara Rubin, Andy Warhol, Carolee Schneeman, Owen Land, Paul Sharits, Michael Snow, Pat O'Neill and Malcolm LeGrice. Arthur goes on to use the terms 'paracinema' and 'performance' to characterise this work, noting that they refer to 'complimentary, and often overlapping, sets of practices' (2005: 4).

In *Between the Black Box and the White Cube*, Andrew V. Uroskie delves in great detail into two examples of what we could call live cinema. Specifically, he explores

Robert Whitman's *Prune.Flat.* (1965), which the artist dubbed a 'theater-piece' (2014: 136), as well as Stan VanDerBeek's *Movie-Mural* (1968), a project Uroskie says 'actively promoted an interpenetration of live and mediated activity' (2014: 156). Further, *Movie-Mural* in particular, through its spatialisation of the projection and its real-time performative capacity 'recast our understanding of spectatorship beyond the traditional model of communicative *reception*' (2014: 169–70; emphasis in original). He continues: 'Rather than broadcasting content for their audiences to receive, VanDerBeek, Cage and Cunningham together sought to orchestrate an environmental situation within which unanticipated forms of communicability and correspondence might spontaneously erupt' (2014: 170). In this way, these early examples of live cinema were more akin to Happenings than to traditional film, and yet, in Uroskie's descriptions, the projects are rife with projections, scrims, screens and media. With his book, and several others exploring these earlier eras of cinematic experimentation, we are finally beginning to comprehend the extent of cinematic experimentation of previous decades.

The Spectatorial Imaginary

Taking hold – literally – of the projector, and by extension, the larger phenomenological attributes of the cinematic experiences, constitutes one gesture for the live cinema artist. Normally occluded, even hidden away in a special room, the projector comes to the fore in this genre of live cinema experimentation. The projection work of Ken Jacobs exemplifies this direction. He often stages live performances in which he manipulates the projector (or multiple projectors) in order to create experiences of what he calls 'deep space'. Jacobs has also experimented with live shadowplay, creating stereoscopic performances using nothing more than stage lights equipped with polarising filters to cast shadows on a rear projection screen. In other cases, Jacobs enlists the audience to participate in transforming their experience from 2D to 3D by shifting a neutral density filter affixed to a wooden chopstick from one eye to the next. For *Opening the 19th Century: 1896* (1990), for example, Jacobs reprinted footage shot in 1896 by the Lumière brothers, such that the disparate original tracking shots now move in a single direction; for the first half of the eleven-minute film, the directionality within the frame is from left to right; in the second half, the directionality shifts so that things move from right to left; to achieve this sense of movement in the frame, some of the shots were placed upside down. For the screening of the project, audience members are instructed when to shift the filter from one eye to the next in order to create the illusion of volume, even though none of the usual apparatus of stereoscopy, such as polarising or anaglyph filters, is present. According to Federico Windhausen, the sense of depth created by this use of simple neutral density filters in conjunction with lateral movement is called the Pulfrich Pendulum Effect. Experimenting with early 3D glasses made for television, Jacobs found that the eye covered with a filter processes signals more slowly than the uncovered eye. 'This mismatch in time,'

explains Windhausen, 'between images that are seen by the naked eye and those that are delayed using the filter, can be exploited to produce depth effects' (2011: np). With this project, Jacobs aimed to recapture what he believes is the sense of wonder felt by viewers in 1896 as they first experienced cinematic motion.

Jacobs also sparks a sense of curiosity and intense perceptual play with his Nervous System projects in which he would pair two analytic projectors capable of sustained freeze frames and single frame advance, each projecting from the same short strip of images; he would then toggle back and forth across the images, frame by frame, playing the images like a DJ to produce a dense flickering dance of movement that seems to extend outward into space. In these live cinema events, cinema and its singular properties are brought to the fore – the projectors are literally positioned in the auditorium rather than sequestered in a sound proof booth – and celebrated in exquisite detail. What we witness as spectators is both the dance of images that seem to hover well in front of the screen, and a deep awareness of the presence of Jacobs, whose embodied effort is more than apparent within the space of the audience.

After 25 years with the dual-projectors of the Nervous System, Jacobs reconceived his apparatus, creating the Nervous Magic Lantern, which he describes as 'strikingly low-tech' (2011: np). It uses a spinning propeller placed in front of a beam of light; Jacobs then puts objects in the path of the light, explaining that we get 'an uncanny 3-dimensional illusion … an illusion of voluminous things moving in great depth as seen from all angles' (ibid.). Jacobs continues to perform with this device, recreating the magic of moving images from more than a century ago.

Similarly, artists Sandra Gibson and Luis Recoder focus on the particularities of cinematic performance, with attention paid to the physicality of light in space. The artists approach their events very much as performances and do not use a written script as a roadmap, preferring instead to understand the event as a 'temporal passagework'. They also focus specifically on the role of the spectator and explain their process in poetic terms. 'To set up the spectatorial imaginary: to assume the role of the one who arrives early, the one who arrives late, in the middle (always in the middle), and who is always already leaving early. To identify with all aspects of this hesitancy that knows nothing but beautiful radical indecision' (in Willis 2008: np).

Recoder's 'Available Light/Liminal Lumen' project aligns both with the work of Minimalism and of the Light and Space artists such as James Turrell, reorienting the viewer's attention from the play of moving images on a single plane to the thick, heavy beam of light that slices through a room when a projector is turned on. Light becomes physical, and viewers begin to question the assumed relationship between viewer and screen, and to the space within which one experiences the artwork. Rather than being sutured into a fantasy story and its image-space, we feel compelled to interrogate the space of the projection, to experience it and feel it in a visceral, tangible way.

Gibson and Recoder also specifically investigate the concept of projection. Writing in a brief essay titled 'Projecting Projection' (2012), they articulate a poetics:

> An aesthetics of projection performance is an apprenticeship to this dedicated custodian of darkness (of nothingness, of disappearance, of invisibility, of transparency). To work or labour in utter darkness is one thing (i.e., to make a bare something out of a bare nothing), but to shape and reshape the intermittent im/palpability of this void through the tyranny of cinematic time is another thing, one which incessantly haunts in its stubborn resistance to resistance. To break free from this temporal tyranny of narrativised time – or at least to slip beneath its gaze – a shift in the projective location presents itself as a possible exit, though by no means an escape. (2012: 61)

For Gibson and Recoder, projection performances are explorations of time and space and these explorations remain confined with the parameters of the cinematic, but the cinematic becomes redefined – or perhaps re-remembered – in a new configuration of light, space, time and presence.

These instances of live cinema mobilise 'liveness' in service to the often occluded role of the projector, which in traditional cinema remains invisible. What's at stake in the revealing the apparatus in this way? First, the projector is recognised as an integral part of the larger assemblage producing an event as a relational experience. Second, the projector, in its previously muffled invisibility, serves as a metonymic reference to all that is hidden in traditional cinematic experience, which relies, even ontologically, on the erasure of its processes. And third, the projector instantiates a particular physicality of the cinematic now lost in the digital era when every computer and smart phone serves as an exhibition space and projectors are increasingly left unused. As such, the projector becomes an emblem – of a particular understanding of time, space and presence; of labour, hidden or exposed; of the machine, the assemblage, the system and that which is normally elided; and of materiality.

While these characteristics suggest what may be at stake in live cinema from the context of the cinematic, Gloria Sutton offers a different explanation of the impetus driving live cinema. In *The Experience Machine: Stan VanDerBeek's Movie-Drome and Expanded Cinema* (2015) Sutton examines the rise of Expanded Cinema live projection performances like those of VanDerBeek, arguing that they were motivated by network-based models of communication. This perspective counters those that trace the evolution of Expanded Cinema primarily within the parameters of avant-garde film and, on occasion, art. In sketching this alternative context and its attention to emerging modes of communication, she writes, 'These include the developments of fiber-optic cable and satellite networks that allowed for the real-time transmission of images over vast geographical distances' (2015: 8), adding that this genealogy is therefore not indebted to technical developments within the context of cinema, nor is it tied to the rhetoric of revolt that characterised the engagement with video during that era. Instead, Sutton aligns Expanded Cinema work with cybernetics and systems theory, and points to a concomitant ethos adopted by these artists characterised by collaboration and transparency. Exposing the projector, then, is part of that commitment to

transparency. Perhaps more significantly, Sutton underscores the union of computation and cinema – a pairing yielding exciting experiments now – in one of its first iterations.

To Construct, Manipulate and Alter Reality

The emphasis on the projector, light and the physicality of the cinematic space is only one characteristic of live cinema. To make a rather large leap, we can trace the lineage of contemporary live cinema practices along other pathways. To take one example, the more musically oriented VJ performances certainly fit the definition of live cinema, but they extend from the realm of live music performance and concert visuals rather than the history of cinematic experimentation. Some of the major VJs of the last twenty years include artists such as Hexstatic, comprised of Stuart Warren Hill and Robin Brunson, who in 2000 released an album entitled *Rewind* as both a CD and CD-ROM, with synchronised videos accompanying each track and tools for remixing the videos. California-based Vello Virkhaus began his VJ career by forming a company dedicated to live visuals called OVT, and later, in 2002, formed V Squared Labs, a company dedicated to creating large-scale concert visuals for major artists including Beyoncé, Coldplay and Mary J. Blige, as well as a major show for Amon Tobin in 2015. Others include D-Fuse, Addictive TV and the Light Surgeons.

Early VJ performances, those in the late 1990s, moved between enhancing live music performances and bringing a sense of performativity to the selection and combination of visual elements akin to the curatorial activity of a DJ with music. In many cases, VJing was seen to be merely illustrative, offering pretty pictures to accompany music. However, VJing can be far more involved. As D-Fuse/Michael Faulkner notes in his book *VJ: Audio-Visual Art and VJ Culture* (2006), VJing has been denigrated through misapprehension:

> Artists view it as eye candy for the clubbing generation; musicians view it as a secondary accompaniment to their music. At best, VJing is regarded as audio-visual wallpaper, not worthy of serious consideration. Yet, to my eyes, the best VJs are creating a new, fluid interface between sound and image – one that is genuinely mould-breaking and aesthetically invigorating, and one that deserves to be recognized as a twenty-first century art form. (2006: 9)

Faulkner's book offers an overview of the artists who are crafting this innovative work. VJing has several defining qualities: it is a live performance, and it involves the creative selection, combination, sampling and/or remixing of existing footage for a live audience, often in collaboration with musicians.[3] VJing also often involves a very direct connection to the music, even digitally transposing audio and video signals or generating imagery that directly references the music. In this capacity, VJing can look to the long history of visual music and filmmakers as divergent as Oskar Fischinger and Len Lye, known for their interest in creating visual experiences of sound.

Yet another lineage could be traced from the work of Jordan Belson, who served as the visual director for the Vortex Concerts at the Morrison Planetarium in San Francisco between 1957 and 1959, where he curated shows using multiple projections to create spatialised experiences of visual music. Belson's interest in consciousness and how it might be explored through film was shared by other artists who contributed to Expanded Cinema and the desire to create an intensive, immersive experience in relation to moving images in the 1960s and 1970s.

Belson's work with the Vortex Concerts was continued in a similar vein in the 1960s with the light shows of the film collective Single Wing Turquoise Bird. The group was known for designing multimedia light shows for bands such as Cream and the Velvet Underground, as well as for more experimental musicians, including Terry Riley and Pauline Oliveros. The members of the group created their shows by combining multiple 16mm projections – between ten or twenty, and sometimes more – to project onto a single screen. The projectionists would also mix coloured chemicals in dishes on the surface of an overhead projector and add gels, or even mask the edges of the projector lens using their hands. When the group used slide projectors, they would move between slides at differing speeds, using dimmers and sometimes producing a strobe effect that could in turn create the illusion of dimensionality.[4]

Contemporary live cinema that is music-oriented also builds on the work of John and James Whitney, who, in the 1960s, experimented with the use of computers in creating multi-screen geometric films based on pattern and repetition in both image and sound. Avant-garde filmmakers such as Norman McLaren and Stan Brakhage are other precursors, thanks to their interrogation of film as a medium, and their experience with single-frame work and the creative possibilities of retinal impact.

Live VJ performances also reference Expanded Cinema experiments of the 1970s, in which filmmakers tried to craft an immersive 'synaesthetic cinema', one that used graphics to create a new kind of vision, which in turn was supposed to lead to a new kind of consciousness. Here, again, we can draw a tenuous connection, linking the desire for the meditative states induced by the abstract computer animations of the Whitneys with live VJ events featuring large-scale, immersive visuals. Former Transmediale curator Jan Rohlf suggests that the interest in the interplay among sound and colour, light and images, and their relationship to consciousness serves at once as a link to the efforts of Belson, the Whitneys and Youngblood, but notes a difference:

> From a general perspective one might argue that with the digital revolution of the 1990s, questions concerning the function of our sensorial organs and the way our psychic apparatus processes and interprets the signals received by our senses gained a new urgency and relevance … This happened mainly because the new digital technologies provide radically enhanced possibilities to construct, manipulate and alter what we consider as our reality. (In Willis 2006: np)

The attention to considerations of reality is extended in the work of artists engaged in generative audio-visual performances in which sounds and images are generated in real time; the idea of a recording, something created and then re-played, as was noted by Douglas Kahn, is replaced by artworks that are emergent, and that do indeed occur 'live', but the notion of 'liveness' is not in the *presentation* of sound in front of an audience, but the *generation* of sound in front of an audience. The work of Davide Quayola, discussed in chapter one, is exemplary here. In addition to his installation work, Quayola has begun designing his own software specifically to generate imagery from sound. Working in collaboration with Natan Sinigaglia and Pedro Mari, Quayola created a real-time visualisation of György Ligeti's sonata for violin solo. *Partitura-Ligeti* (2012) visualises the sound of a viola as it is transformed into a dynamic, graphic score. One of a series of six pieces that all translate music into abstract geometries and forms, the piece transposes the long, lingering tones made by the stringed instruments into a series of intertwining, coloured lines that writhe and warp in rhythm with the sound. Quayola has said that he is interested in creating his own visual language in this work, and he does so, moving within a history that includes Oskar Fischinger's life-long quest to visualise music; here, however, the music is not represented visually but produced visually.[5] The sound produces its own images.

For Japanese artist Ryoji Ikeda, sound also serves as the foundation for visuals. For example, in a live performance called *Test Pattern* (2008), sound is converted directly into black-and-white patterns projected on a very large screen. The project is part of the artist's series of pieces called 'datamatics', which explore the visualisation of the invisible data that surrounds us. *Test Pattern* is part of a 'system that converts any type of data (text, sounds, photos and movies) into barcode patterns and binary patterns of 0s and 1s. The project aims to examine the relationship between critical points of device performance and the threshold of human perception, pushing both to their absolute limits' (Ikeda 2008: np). The goal of *Test Pattern,* then, is to constitute a machine for generating affect.

While interest in perceptual experimentation and formal correlations between sound and image have motivated an eclectic range of live cinema work, other artists have focused on the potential for live performance to explore strategies of visual reinterpretation that are expressly political. Building on the rich history of appropriation art and culture jamming, live cinema performances such as DJ Spooky's remix of D.W. Griffith's *Birth of a Nation* (1915) seek to highlight the overt racism of the original film, critiquing it through fragmentation, repetition and densely mediated digital effects aimed at a politics of reversal. Like Oscar Micheaux's *Within Our Gates* (1919), which offered a narrative response to Griffith's epic that resonated with the racial volatility of the post-World War I era, DJ Spooky's *Rebirth of a Nation*, which he began to perform in 2004, appropriates and recontextualises whole sequences from Griffith's film in order to critique them against the backdrop of continuing racial tensions in the U.S. nearly a century later. Originally commissioned for a series of international media festivals and performed live more than fifty times around the world, *Rebirth of*

a Nation was re-released on DVD with music performed by the Kronos Quartet in 2015, again resonating with public protests against racially motivated violence and the #BlackLivesMatter movement.

Also mixing cultural politics with strategies of appropriation, VJ Um Amel (Laila Shereen Sakr) created a series of live cinema performances during and in the immediate aftermath of the 'Arab Spring' uprisings of 2010–12. Sakr's performances are characterised by rapid-fire and densely-layered visuals depicting events and iconic images emerging from the revolts across the Middle East, while also drawing on the artist's unique English and Arabic language repository of social media and analytics, R-Shief. In her *Egyptian Body Politic: Adaptation of #Tahrir* (2011), for example, VJ Um Amel mixes her own voice with music and sound layered over YouTube videos, animations, and a live video feed, as well as text and analytics from millions of Tweets revealing patterns of social media mapped onto these historic events.

While VJ culture, with its potential to explore both formal and political registers, represents the evolution of one form of live cinema, I turn now to the work of artists who investigate a different set of configurations of the cinematic, explicitly reimagining the apparatus of 'cinema' in order to experiment with altered forms of interpellation, knowing and being within a networked culture.

Mia Makela: Live Cinema Is Not Cinema

Mia Makela, a Finnish artist who performs under the name Solu, is one of several international visual innovators dedicated to live cinema with attention to the ways in which the cinematic experience becomes a performative event unfolding in real time. Makela's work is often moody, ethereal and dream-like, presenting cinema as a space of the imaginary, coming before narrative and language to enact experiences of affect and the sensorium instead of logic and story. Her style ranges from minimal abstractions to multi-layered compositions in dreamlike narrative journeys.

Makela began performing with visuals in 2001 at festivals around the world, and she has written and lectured on the history and forms of live cinema internationally. She bases her practice on her love of editing, which is not in service to a story but rather oriented toward the crafting of an entire environment. In 'The Practice of Live Cinema', Makela writes that 'live cinema is not cinema' at all (2008: 1), suggesting instead that live cinema is formed in the ways in which an artist choreographs transitions between sequences of story. David Lynch's *Lost Highway* (1997) serves as a reference point, with its eerie shots of the highway at night, the asphalt stretching out in front into the dark. According to Makela, these shots 'are the basic material for live cinema performances: the transitions, the movements, the pure visual beauty and intrigue, the atmosphere' (ibid.). Makela goes on to list the components necessary to constitute live cinema, namely projection, the performer or creator, an audience, and the space and time of the performance. More specifically, according to Makela, live cinema acknowledges projection, often calling attention to the placement of the

projector – or multiple projectors – and the materiality of the technology and the light. Live cinema also involves a performer, or performers, who participate in the event, contributing another element of the performative. The audience is vital for live cinema, and artists engage the space between performance and audience in radically different ways. Finally, the spatial and temporal elements of live cinema also vary depending on the performer and venue.

In her parsing of the space of live cinema, Makela distinguishes among digital spaces generally; the space of the desktop and the interface to the software that the performer engages intimately during the performance; the performance space; the projection space, which, in leaving behind the requirement of traditional cinema's flat-screen explores new arrangements of audio-visual experience; and the physical space, which for Makela designates the shared space of audience and performer. In terms of parsing the temporal aspects of live cinema, Makela is less clear, but makes a key point that what we understand to be 'real time' is inexact at best; the 'live' in live cinema can reference the fact that real time images are produced and received simultaneously. For Makela, this can mean generating visuals in real time, or even using a live camera as part of the input for a performance.

In her attention to the specifics of live cinema, Makela has made a major contribution to the field. Her performances further this work, helping enact her ideas in tangible experience. With the performance of *Kaamos* (2007–8), for example, we enter a dream space of sunlight, moonlight, water and the passage of seasons. Images pulse and disappear; we see glimpses and small circles of imagery as we move through a forest. There is rain, and a sense of stuttering beauty aligned with the music as we follow a figure on a journey. Makela offers an experience of narrative from which the narrative has almost been completely dissolved, leaving us not with the memory of a story so much as with an experience of an emotional field.

For the performance of *Kaamos*, however, Makela is nearly invisible, and it is difficult for an audience to know what constitutes the 'liveness' of her performance. Indeed, viewers are sometimes completely unaware that the visuals are being mixed in real time; the event is understood – or rather misunderstood – to be the projection of an experimental film rather than the unfolding of a live performance. In this way, one of the crucial aspects of the performance is elided. While Makela is certainly correct in underscoring the ways in which our understanding of 'live' and 'real time' shift, it is precisely at this nexus that the post-cinematic and posthuman reside: what indeed is 'live'? And why at this particular moment do we care?

Layering Realities: Miwa Matreyek and Cloud Eye Control

Los Angeles-based media artist Miwa Matreyek begins to answer this question with her work, which blends elements of theatre, performance, animation and cinema to create live events that layer multiple projections, recorded music and the image or silhouette of her own body moving through the projected images (see Koch, Pantenbug

and Rothöhler 2012). The result is a powerful experience for an audience as we simultaneously marvel at the enchanting animated worlds she creates and attempt to imagine how the entire performance is functioning. Matreyek specifically catalyses a series of questions for the audience: What is real? What is animated? What is projected? What is shadow?

Matreyek's work is dedicated both to the formal and conceptual possibilities of animation and collage, as well as to breaking down the languages of diverse art forms. Nearly a decade ago, while a student at California Institute of the Arts, Matreyek began a collaboration with fellow student, Chi-wang Yang, exploring among other things the potentials of front and rear screen projection. They named their collaboration Cloud Eye Control and were later joined by Anna Oxygen. As a team, they began designing live performances that united their overlapping skills in interactive multimedia, projection, animation, musical composition and performance. One of their early pieces is *Ocean Flight* (2005), which includes video footage of a person running; the footage is captured live during the performance on one side of the stage, but then projected onto another screen on the opposite side of the stage, where it is collaged together with yet another projection and another live performer. The mix of projection and live performance produces a dizzying and captivating kind of vertigo as one tries to unravel the relations between live-action and projected imagery. The piece creates an image machine within which the performers are thoroughly enmeshed; the boundaries between human and backdrop give way to an undifferentiated field in which performer and image merge. Commenting on *Flight Pattern,* Yang says, 'In some ways, this could be done so much better through some sort of green screen and a real-time compositing effect. But the intention is very much to physicalize media and to actually start to layer and give body, depth and live presence to media and the body' (in Willis 2015: np). He continues:

> The intention was to try to create a circuit, a process so you see the camera, you see the computer station, then you see the camera capturing the body, and then you see that body showing up on the screen in front of you, and then you see that body being layered in front of media, so the idea is to create a whole traversal, a virtuosic transmission or movement of the body. It seems so pedestrian now to travel across the ocean, so the idea is to show that what we now think of as spectacular transversal transportations are virtual ones through networks and mediation. (Ibid.)

Here Yang clearly articulates the larger conceptual agenda of the group, with its desire to enact the intersections of lived bodies and networked systems.

Matreyek, in her own work, also extends these ideas. Her MFA thesis project is entitled *Dreaming of Lucid Living* (2007). In this piece, the artist explores sleight of hand and specifically attempts to create a continuum between the virtual and the real. The project combines playful and yet visceral animated collages – ornate drawings of skeletons, for example, that seem to come from a children's book – which are projected.

Matreyek then layers the silhouette of her own body within the projections. She also creates entirely black-and-white images such that her own silhouette can mingle among the projected images and the viewer simply cannot tell what is the shadow of the artist and what has been previously animated. In all of her work, Matreyek moves in carefully choreographed synchronisation with the animations, becoming part of the machine of the cinematic event she is creating.

One of Matreyek's more recent pieces is *This World Made Itself* (2013). Like the other works noted here, the project integrates projected imagery and performance. It begins with animated images of molten rock and the formation of oceans and landmasses, and then moves on to the origins of life. Richly detailed and colourful – and yet entirely surreal – the imagery looks as if it were borrowed from a children's encyclopaedia from the 1950s. Matreyek's silhouette once again intercedes in the projected imagery, creating a mythic female presence moving gracefully through the prehistoric scenes. We see her swimming in an ocean of fire, walking through tall grasses, traipsing across mountains.

In Matreyek's work, the performances create a doubled story and the same can be said of Cloud Eye Control's projects as a whole. Matreyek explains: 'We like to say that there are the two narratives. There is the story being told, cinematically, as a composition on the screen and around the screen, but there is also the narrative of the staging and the theatricality, and actually seeing how these things come together to make the cinematic space' (ibid.). Viewers engage with a story that unfolds on-screen, but at the same time we are highly attuned to how that narrative is being staged, how the images are being made, and how meaning hovers between the representational narrative and its enactment. We are caught between these two realms, but rather than experience a sense of frustration or alienation, the duality is pleasurable.

This pleasure, I would argue, is a pleasure of both the post-cinematic and the posthuman. Matreyek's work rejects the production of the viewing subject of classical Hollywood cinema in which highly codified forms of visual design, mise-en-scène and editing craft a fictional coherence, not just of time and space, but of the viewing subject in relation to a protagonist. This viewing subject aligns with the subject of modernity, which presumes that the subject is whole, coherent, unified and fully engaged, even produced, by the narrative. However, in Matreyek's work, pleasure does not occur in relation to the suturing of the audience member into an on-screen narrative. Instead, we are engaged in a process of interpretation and an attempt to understand the assemblage producing the experience we witness. We are no longer content with merely a story; instead, we query the machine, and the ways in which human and non-human interact, and we take pleasure – though perhaps anxious pleasure – in the blurring of boundaries.

If the projection and narrative form two points of enquiry, so, too, does the artist's body, which becomes ephemeral in its role as silhouette and shadow. While classical Hollywood cinema tends to produce both a coherent protagonist and a clear mode of identification, Matreyek's work instead presents a body with blurred boundaries, one

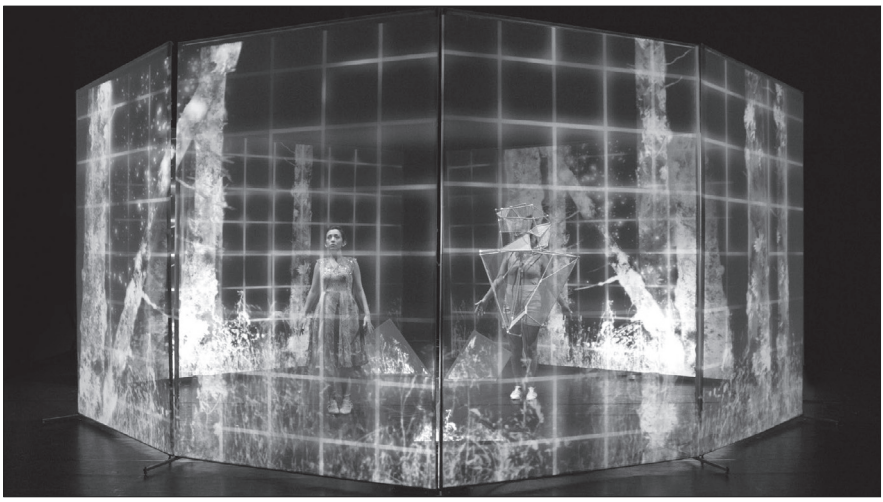
often entirely coincident with the mediated environment within which it is situated. This is a body – and a subject – that is produced in relation to the world; the self is not an isolated entity, but an ongoing conjuring with its environment. And indeed, Matreyek does little to hide the fact that she is behind the screen. In some instances, it is possible for viewers situated to the left or right of the centre of the screening space to glimpse evidence of her presence. Once again, the pleasure is not in being fully tricked, but rather in the oscillation between knowing and wondering, and in witnessing the dissolution of the body's boundaries as it melds with the mediated world.

Returning to the work of Cloud Eye Control, in 2009, the trio created a fantastical live cinema work called *Under Polaris*, which brings together animation, live music, projection and performance to tell the story of a woman's journey to the North Pole in order to deposit a seed carrying the essence of humanity in the deep ice where it will remain safe. With this project, Cloud Eye Control began to work at a larger scale, creating a piece that is at once spectacular and mythic in terms of its narrative, while also highly controlled in its staging, and yet at the same time cheerfully handmade. Indeed, the performance boasts what artist Jem Cohen calls a kind of 'primitive enchantment' (in Lim 2013: np), referring to the simple magic of light, shadow and the illusions that are possible with these basic elements.

Overall, the project helped continue to refine the working vocabulary for the trio of artists as they navigate among three very different art forms, namely animation, theatre and music. It also was large enough in terms of scope to allow them to more fully engage with the design of an entire world with multiple levels of reality, and to explore with more complexity the role of the body in relation to the mediated environment. As Yang explains, 'We often have multiple layers of screens on stage, and also multiple layerings of densities in order to create multiple first plane, second plane, and third plane layers' (in Willis 2015: np). Yang notes that the layering of screens replicates the live compositing effects that are possible, creating illusions that are very powerful, as when the protagonist is paddling across the ocean:

> Also, by not ever really using a terminal surface like a screen in the cinematic paradigm, the image is always put in relationship to a body on stage. Even when the screen is placed in the background, it is never the final, terminal surface that we are accustomed to in the cinema. We are always putting the body in relationship to images in space. (Ibid.)

In their introduction to *Performance and Technology: Practices of Virtual Embodiment and Interactivity* (2006), Susan Broadhurst and Josephine Machon note that the two quintessential elements in mediated performance are the digital and the corporeal. They go on to argue that these two foci demand a new mode of analysis and interpretation, one that 'foregrounds and celebrates the inherent tensions between the physical and the virtual' (2006: xvi). Further, they point to 'the immediacy of the physical/virtual body' and understand this immediacy to be vital (ibid.). The work of Miwa Matreyek



Half Life (2015) by Cloud Eye Control; image by Eugene Ahn

and Cloud Eye Control explores this tension between the physical and virtual, most often through the body and its reconfiguration through media. The body is at once entirely physical, but also subject to projection and reconfiguration via the media with which it interacts.

The imbrication of body and technology continues full force in Cloud Eye Control's *Half Life* (2015), their most ambitious project to date. The story emerged from reports following the 2011 Fukushima Daiichi nuclear crisis and is rife with anxiety about ecological disaster and a world in turmoil. The opening image shows two buildings side by side; satellites dot the rooftops and, on occasion, a shade within a window rises or lowers. Viewing the image prior to the start of the performance, it is difficult to know the status of the image – still or moving? Layered or flat? As the performance begins, the screens – four of them – rise. A woman appears, and, for the remainder of the performance, we will watch her negotiate a series of spatial relationships with the world around her, some large, some small. A swirl of paper and trash cavorts around her as she walks through a dystopic wasteland; meteorological mayhem with lightning tosses her backwards; she is doused in rainwater while standing in a fantastical forest; and she is nipped by fish swimming past her as she wades in the ocean. Each of these image sequences melds the human body with an animated world, suggesting a new definition of animation with reference to its etymological root to 'instil with life'. Other sequences, however, veer far from the natural world to the technologised milieu, placing the woman within a matrix of measurement, statistical analysis and control. This imagery is data visualisation, and the body is integrated into its systemic reduction. Matreyek, explaining the larger mise-en-scène of the performance, says, 'We have basically created a cinematic environment that the performer is inside of. In other scenes, though, the screen is treated like a surface in the domestic space, like a big

user interface' (in Willis 2015: np). This interface is represented graphically, but it also includes scenes when footage is captured that appears to be web-cam images. In one particularly visceral sequence, the woman discovers something on her face. Looking closer, which the audience sees as two giant projections of real-time image capture, she peers at her cheeks and chin, which begin to distort, becoming disfigured, discoloured and horrible. The image poignantly captures a desperate sense of anxiety, but the sequence gains its intensity precisely through the familiarity of the video image.

Matreyek continues to explore the rendering of space in the group's performances:

> We are constantly recomposing the stage to create new relationships between the body of the performer, the space and the projections. Some moments are staged in a more a panoramic environment, to show that she is kind of lost in this world, as opposed to other scenes, in which there are ways that the screens compose a room or container that the performer is inside of; they become her psyche. (Ibid.)

In this way, Cloud Eye Control pays specific attention to the relationship between the body and the various screens and projections that constitute the story world. What they render, then, is a new habitus, to borrow from the terminology of Pierre Bourdieu, who understood the habitus to be a set of practices and dispositions that are at once deeply social and embodied. The world of *Half Life*, with its differing layers of spaces and screens, provides a depiction of a world in which to be embodied is to be enmeshed; to be embodied is to be is to be screened.

Cinematic Entanglements: The Work of Mariano Pensotti

While Miwa Matreyek and Cloud Eye Control render a vision of the body in relationship to a mediated world, Mariano Pensotti, an Argentinian artist whose work combines cinema, visual art and theatre, considers the post-cinematic, posthuman concept of identity. His projects include *Sometimes I Think, I Can See You* (2010) in which four writers with laptops sat in a train station writing words that appeared on four screens in the station. Passers-by became part of the narrative, at once performing within the fictions created by the writers and witnessing their reconfiguration on the screens. Much of Pensotti's work explores this intersection, considering the connections among disparate media forms, merging text and image, theatre and cinema.

Pensotti's *Cineastas* (2013) is a theatrical work with elements of the cinematic that tells the story of four filmmakers and their individual movies, focusing on the intersection of personal lives and the stories the filmmakers desire to tell. The project is set within a two-tiered stage, with the 'real' lives unfolding on the lower level, and the often more lurid cinematic stories taking place on the upper level, with bright lighting and sound effects. What makes the project compelling in the context of the posthuman and the post-cinematic are the ways in which it adopts the elements of cinematic language – split screen, voice-over, temporal elisions – and the manner in

which the characters – there are five – seamlessly transition from body to body, each actor adopting different roles as the story unfolds.

The primary thematic concern for *Cineastas* is the relationship between fiction and reality, with the director asking on the project's website, 'Do our fictions reflect the world, or is the world a distorted projection of our fictions?' How, he wonders, do the stories the filmmakers tell through their films impact their lives? And how do the specifics of their everyday lives affect their films? We might extend his enquiry to consider the ways in which the structures of storytelling affect our understanding of self, knowing and being. How does the transition to a form of theatrical/cinematic experimentation that eschews unified characters and coherent meaning to produce something more akin to a machine of storytelling contribute to the elaboration of the posthuman experience?

Cineastas juxtaposes the so-called real and the fictional, with an on-stage narrator helping bridge the gaps, and the actors taking turns playing the diverse roles needed to complete what are essentially eight different stories. These eight stories become reconfigurable assemblages, churning through segments of narrative; and the bodies of the actors become figures within a constantly unfolding environment in which character identity is transient at best. The layerings of fiction and reality give way to simply a layering; we come to experience not a story, or even eight stories, but the machinery of experience and the production of meaning as an ongoing enactment choreographed among an array of selves and the world. The world is not merely inhabited, with the space serving as inert backdrop; instead, the world and the characters both are produced through their interactions. Indeed, in place of an unfolding story, *Cineastas* presents a richly textured tableau or even a backdrop where experience unfolds through the interactions of actors and space. Despite the fixity of the run-time of the performance and the physical confines of the set, temporal and spatial continuity give way. In their place is a sense of profusion and possibility.

The structure of the narrative also points to the posthuman and post-cinematic in its recursive embedding of stories within stories. As the eight narratives unfold, they mix and merge, not so much in a hierarchical embedding that would move from one plane upward or downward to another plane. Instead, there is a sense that the levels exist interchangeably on the same plane. The history of embedded narrative is rich and much theorised. Tzvetan Todorov, for example, writes, 'Embedding is an articulation of the most essential property of all narrative. […] For the embedding narrative is the narrative of a narrative. By telling the story of another narrative, the first narrative achieves its fundamental theme and at the same time is reflected in this image of itself' (quoted in Clarke 2008: 97). Similarly, Italo Calvino finds the layering of narrative to be constitutive of narrative itself: 'Literature does not recognize Reality as such, only *levels*' (1987: 120; emphasis in original).

This layering of levels and the fluidity of characters as the actors play diverse roles contribute to a sense of ontological flux. Characters are not anchored to bodies; they morph and return. It becomes impossible to engage the project through a process of

character identification, or to contain the stories within a cohesive whole. Instead, we remain at a distance, lacking the privileged position in classical filmmaking, as well as all the assurances of unity and coherence once promised. *Cineastas* disrupts the centrality of character identification by blurring the boundaries of character identity, both conceptually and physically.

Describing posthuman subjectivity, Tamar Sharon writes:

> Modernist thought assumes that the body is the unambiguous locus of the self (agency, consciousness, mind), the ground of identity, a vessel occupied by and at the disposal of an animating, wilful subjectivity. […] It is specifically within a bounded, molar body-as-organism that the modern subject is located because it acts as the boundary, limit, edge or border of subjectivity, that which divides the subject both from other subjects and from objects in the world. (2014: 136)

Similarly, in describing affect and bodies, Lisa Blackman writes, 'Rather than talk of bodies, we might instead talk of brain-body-world entanglements, and where, how and whether we should attempt to draw boundaries between the human and non-human, self and other, and material and non-material' (2012: 1). The use of bodies in *Cineastas* releases the subject from the single body, creating something much more akin to Blackman's brain-body-world entanglements as the boundaries among humans, and those between human and non-human slip and merge. The individual body gives way as the nexus of subjectivity; in place of the bounded modern subject, there is instead divestiture and dispersal.

From Narrative Pleasure to Parallax Play: The Work of Janie Geiser

Like Miwa Matreyek and Mariano Pensotti, artist Janie Geiser disrupts the boundaries among media forms, preferring to mix and match to create hybrid works dedicated to conjuring atypical experiences of story and engagement that, like the work discussed above, eschews traditional forms of pleasure, identification and unity in favour of a form of posthuman subjectivity. Known both for her work in puppet performance and experimental filmmaking, Geiser's live cinema work explores the melding of performance, puppetry and projection, and in the process, contributes to the emergence of a new intermedia language.

Clouded Sulphur (death is a knot undone) (2013) is a hybrid puppet performance project with live video projection that borrows from *Bunraku*, a form of traditional Japanese puppet theatre. Specifically, Geiser was inspired by the seventeenth-century playwright Chikamatsu Monzaemon, who was the first to use the dramas of everyday people, rather than those of emperors or elites, as the foundation for his puppet theatre. His *The Love Suicides at Sonezaki*, for example, was based on a real incident, and became a tremendous success among audiences. Inspired by Chikamatsu's attention to the everyday, Geiser conducted research, looking for a real event that could form the foun-

dation for her own puppet theatre work, one that would perhaps resonate in a way akin to those of Chikamatsu for contemporary audiences. She selected the tragic tale of young Brenda Sierra, and the resulting piece explores the kidnapping and murder of the fifteen-year-old East LA sophomore, who left her home for school one October morning in 2002 only to disappear completely; her body was found in the dusty hills of Crestline near the San Bernardino Mountains a few days later, her death prosaically attributed to a blow to the head. While there is speculation about why she might have been murdered, no one has been indicted in the case.

Set in a darkened theatre space, the action of *Clouded Sulphur* takes place on a large hillside that slopes downward and fills most of the small theatre space across one horizontal side of the room. The hillside set is raised to waist level, on a platform, and two narrators sit on another raised platform to the right of the stage, while the musician performs on yet another elevated scaffold to the left of the stage. The small audience sits in several long rows along the opposite side of the long wall, so that the experience overall is extremely intimate, as if you are sharing the space of death and loss with the puppets and their handlers.

The puppets have carefully crafted wooden faces with wide eyes and angular noses, and arms, legs, hands and feet that are hinged to allow very nuanced, delicate movements. There are three characters: Brenda, her sister Fabiola and her brother Julio; Brenda wears a red hoodie that matches the sweatshirt worn by the real girl when she was killed. Each puppet is handled by three people who, working with extreme precision, are able to produce strikingly subtle and lifelike movements in the figures. The puppets do not seem human necessarily, but it is difficult not to experience them as utterly alive given the nuance of their movements. As Brenda crawls slowly backwards down the hillside in confusion, or as Fabiola collapses to her knees to claw feverishly into the dirt with her bare hands, you want to reach out and hold them, to offer some solace and protection. Barring that, you simply want to weep.

Helping create the overall environment, one that does not merely tell a story but creates an immersive experience, the narration – spoken by the narrator – consists of elusive fragments, a poetic patchwork of details and history, while the music crafts an elegiac mood that is haunting without being overbearing (the piece was written by playwright Erik Ehn and the music composed by Valerie Opielski). Projections showing a live feed from a camera trained on a detail of the larger performance appear on the landscape, magnifying certain key moments or offering further texture and emotion.

Within the matrix of performers, images, music and narration, the human is decentred or displaced by the wooden puppets, who form the centre of attention, but this centre is at all times a nexus. We watch the figures, and they are lit to be viewed, while their handlers, dressed in black, remain in the edges of shadows. Perhaps most importantly, voice is severed from bodies; the puppets do not speak, nor do their handlers. The narrator speaks from across the room and above.

Writing about the *Bunraku* form in *Empire of Signs*, Roland Barthes highlights this separation. *Bunraku*, he writes, 'shows the gesture, lets the action be seen, exhibits

simultaneously the art and the labour, reserving for each its own writing' (1983: 54). He adds that this showcasing of both the art and the labour achieves the distanciation recommended by Bertolt Brecht. Barthes continues, noting that the 'distance is made explicable by *Bunraku,* which allows us to see how it can function: by the discontinuity of codes, by this caesura imposed on the various features or representation, so that the copy elaborated on stage is not destroyed but somehow broken from within (ibid.). This 'discontinuity of codes' and the 'caesura' signal a key component of the posthuman, namely the experience of viewing the entire storytelling system as system. The distanciation affords an awareness of not merely the main character and her experience, but the entire world of the story, from the puppet handlers and their expressive attention to the bodies of the puppets, to the narrator, musicians and projections, all of which contribute to telling the story.

Geiser's next puppet-based performance was *Fugitive Time* (2014), which is composed of multiple moving planes, video projections, shadows and disparate times and places to explore the story of a woman with tuberculosis who is moved to a sanatorium. The space of this project is far more mutable than that of *Clouded Sulphur*, and makes use of rolling scrim panels akin to those seen in hospital rooms; they are moved throughout the performance to reconfigure the space as a constantly shifting environment. In addition to forming walls, the screens also function as surfaces for video projections, which show imagery of Los Angeles, hills and trees, as well as archival photos of the tents and cottages that were part of an LA sanatorium from the past. In this case, the projections are much more elaborate than in the earlier piece, creating

Fugitive Time (opening scene) (2014-15) by Janie Geiser; image: Vincent Richards

momentary fictional spaces, travel scenes and evocations of landscape. As such, the projections conjure the cinematic, producing new times and spaces. However, they never engage spectators as traditional film viewers; instead, they produce a series of transitory story worlds that come and go.

Both *Clouded Sulphur* and *Fugitive Time* bring together many different media elements – beguiling puppets, elaborate sets, projected video, evocative lighting, sumptuous music and live performance – to create experiences that are themselves between worlds. Neither just live theatre, puppet show, video screening or story, the projects cross boundaries and invite audiences to experience intense moments of identification and emotion in formats that are unfamiliar and unpredictable.

Traditional identification is also disrupted. Rather than connecting with recognisable characters as we do in a film or play, we work to identify with wooden faces and lanky puppet bodies. Rather than enjoying the predictable rhythms of the conflict-driven three-act structure, we cobble together bits and pieces of narrative to make the stories whole. Rather than settling back to watch a feature film or play, we lean forward, working to make sense, to feel, to know. It is not the often spectacular merging of disparate elements that explains the power of Geiser's work, and it is not the ardent pathos of the stories she chooses to tell. Instead, Geiser leverages all these different elements into a new arrangement, and what she achieves is a new form of collective thinking and experience, one that seems perfectly suited to our current moment.

In Geiser's work, and live cinema projects that engage specifically with the post-cinematic, we witness a shift from narrative pleasure to parallax play. The traditional pleasure of immersion within a singular, all-engaging story world gives way to the play involved in viewing the entirety of the storytelling assemblage. No longer are we swept away into another world; instead, we exist in a parallax experience of witnessing not just the story but also exploring its edges and mechanics. Rather than finding pleasure only within the story, we now enjoy understanding how the story is produced. Rather than suspending disbelief, we enjoy an awareness of suspension itself. In other words, rather than cheerfully acquiescing to the story and its conditions, we enjoy precisely the suspension between real and fictional, world and model, due in part to the immediacy of what we are witnessing. Writing about contemporary performance in an essay entitled 'Digital Culture', Andy Lavender describes 'the simultaneous coexistence, the mutual play of what might appear to be two distinct media – the screen and the stage – and the ways in which their very co-relation produces effects of immediacy that are deeply involving – more, deeply pleasurable – for spectators' (2010: 56).

In the work of Mia Makela, Cloud Eye Control, Mariano Pensotti and Janie Geiser, the singular protagonist that dominated the classical narrative film is displaced by the environment, the system and the assemblage, and we are called on to reckon with all parts equally. Ralf Remshardt addresses the centrality of the system: 'In a post-human performance paradigm, spectator and performer both relinquish their positionally determinate (dialectical) claims to presence and reconfigure themselves as dynamic, interdependent parts of an emergent system' (2010: 135). The system includes

performers and spectators, but also projections, screens, objects, moving sets and other material entities, all of it harnessed into a system designed to celebrate relational interconnectivity. As Remshardt writes: 'Posthuman refuses to close down the available connections, intersections, and nodes; rather it insists on making them visible and articulating the need to (re) connect with them' (ibid.).

Jan Rohlf, in discussing the futures of live cinema, eloquently imagines the merging of once separate entities:

> When artist and machine are short-circuited as unstable systems, when sound is output as light and specific qualities of this light are fed back into the sound-generating process by the artist, when performer and machine interact, when the artists' decisions are confronted with autonomous generative processes controlled by the machine: with all of this we get an idea of how man and machine find new modes of perception that are only possible through our interaction with the technical medium. Live cinema performances thus contain relevant questions about the subject's relation to and its role as an integral part of its technical surrounding. (In Willis 2006: np)

The post-cinematic and the posthuman: together, each completes the project of the other, attempting to reckon with experience, identity and subjectivity in a networked culture. As cinema is reinvented as an intermedia form, and as we come to terms with a world in which the human is no longer the centre, artists contribute much to a conjuring of new arrangements, attunements and practices of being through works that call us forth in new ways.

making, for example, Goodman writes, 'The arts must be taken no less seriously than the sciences as modes of discovery, creation, and enlargement of knowledge in the broad sense of advancement of the understanding, and thus that the philosophy of art should be conceived as an integral part of metaphysics and epistemology' (1978: 102). He also articulates his sense of the significance of world making: 'Works of art can participate in world making precisely because they have symbolic functions' (ibid.).

4 See Virginia Pitts, 'Writing From the Body: Kinesthetics and Entrainment in Collaborative Screenplay Development' (2013), for a discussion of the approach to embodied collaborative processes.

Chapter Three: Live Cinema

1 See Michael Lew (2004) 'Live Cinema: Designing and Instrument for Cinema Editing as a Live Performance', for another definition, and David Fodel (nd), 'Live Cinema: Context and "Liveness"'.

2 See, also, Steven Ball, David Curtis, Duncan White and A. L. Rees (eds) (2011) *Art, Performance, Film*; Andrew V. Uroskie (2014) *Between the Black Box and the White Cube: Expanded Cinema and Postwar Art*; and Gloria Sutton (2015) *The Experience Machine: Stan VanDerBeek's Movie-Drome and Expanded Cinema*.

3 See Timothy Jaeger (2006) *Live Cinema Unraveled: Handbook for Live Visual Performance,* in which he lists a series of 'codes' that characterise the form, including mixing, sampling, the use of filters and spatial montage within the frame.

4 For more on Single Wing Turquoise Bird, see David E. James (2005) 'Expanded Cinema in Los Angeles: The Single Wing Turquoise Bird'.

5 See William Moritz (2004) *Optical Poetry: The Life and Work of Oskar Fischinger* and Cindy Keefer and Jaap Gundemond (2013) *Oskar Fischinger: 1900–1967* for more on the work of Fischinger.

Chapter Four: Urban Screens / Screened Urbanism

1 See Erkki Huhtamo, 'Messages on the Wall: An Archaeology of Public Media Displays' (2009), for a history of early billboards and urban signage.

2 See Anne Friedberg's 'Urban Mobility and Cinematic Visuality: The Screens of Los Angeles – Endless Cinema or Private Telematics' which explores 'the tension between the material reality of built space and the dematerialized imaginary that the cinema always provided' (2002: 186).

3 See Rita Raley's *Tactical Media* (2009) for a discussion of political and activist media practices in public space.

4 For more information on video art in public space, see Catrien Schreuder's *Pixels and Places: Video Art in Public Space* (2010); for more on installations in public space, see Erika Suderburg's *Space, Site, Intervention: Situating Installation Art* (2000) and *Art Works: Place,* by Tacita Dean and Jeremy Millar (2005).